

Lesson 22: Fourier Transforms, Power Spectra & K-Space

Professor Dave Explains Differential Equations Master Series · Lesson 22

Video: Fourier Series and Transforms Part 2: Power Spectra and K-Space Course: Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that while Fourier Series decompose repetitive periodic signals with discrete harmonic spikes, real life features non-repeating pulses: a single clap of thunder, an optical laser burst, or a transient seismic jolt. By letting the period stretch to infinity ($T \rightarrow \infty$), the discrete frequency spikes blend into a smooth, continuous spectrum: the Fourier Transform. Dave also demystifies 'K-Space'—the spatial frequency realm where X-ray crystallographers and MRI machines view atoms before taking an inverse Fourier transform to construct a physical picture.

2. Core Theoretical Framework & Rigorous Formulations

1. The Continuous Fourier Transform Pair

$$\hat{f}(\omega) = \mathcal{F}\{f(t)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt$$
$$f(t) = \mathcal{F}^{-1}\{\hat{f}(\omega)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\omega)e^{i\omega t} d\omega$$

2. Parseval's Theorem & Power Spectral Density

Total energy in the time domain equals total energy in the frequency domain (Energy Conservation):

$$\int_{-\infty}^{\infty} |f(t)|^2 dt = \int_{-\infty}^{\infty} |\hat{f}(\omega)|^2 d\omega$$

$S(\omega) = |\hat{f}(\omega)|^2$ is the **Power Spectral Density (PSD)**.

3. Spatial Frequency & K-Space

Replacing time t with spatial position $\mathbf{r}$ yields spatial frequencies $\mathbf{k}$ ($k = 2\pi/\lambda$). In MRI and X-ray diffraction, raw signals are gathered in reciprocal K-space, then reconstructed into physical tissue scans via the 2D Inverse Fourier Transform.

3. Fully Solved Whiteboard Problems

Physics Problem: Fourier Transform of a Rectangular Gate Pulse

Problem Statement: Compute the Fourier transform of a rectangular pulse of duration $2a$ and unit height: $f(t) = \begin{cases} 1, & |t| < a \\ 0, & |t| > a \end{cases}$.

Step 1: Apply Direct Transform Integral

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-a}^a (1)e^{-i\omega t} dt.$$

Step 2: Evaluate Exponential Integration

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-i\omega t}}{-i\omega} \right]_{-a}^a = \frac{1}{\sqrt{2\pi}} \left(\frac{e^{-i\omega a} - e^{i\omega a}}{-i\omega} \right) = \frac{1}{\sqrt{2\pi}} \left(\frac{e^{i\omega a} - e^{-i\omega a}}{i\omega} \right).$$

Step 3: Convert to Trigonometric Form via Euler's Identity

$$\text{Recall } \sin(\omega a) = \frac{e^{i\omega a} - e^{-i\omega a}}{2i}. \text{ Therefore: } \frac{e^{i\omega a} - e^{-i\omega a}}{i\omega} = \frac{2 \sin(\omega a)}{\omega}.$$

Step 4: Express in Canonical Sinc Form

$$\hat{f}(\omega) = \sqrt{\frac{2}{\pi}} \frac{\sin(\omega a)}{\omega} = \sqrt{\frac{2}{\pi}} a \operatorname{sinc}(\omega a).$$

Step 5: Physical Wave Interpretation

A sharp localized rectangular box in time transforms into an oscillating, decaying sinc function in frequency space. Narrower time pulses ($a \rightarrow 0$) require an infinitely wide spectrum of frequencies—the exact mathematical bedrock of the Heisenberg Uncertainty Principle!

⚠ Common Exam Pitfall & Trapdoor Avoidance

Notice that when normalizing the Fourier transform, constants like $1/\sqrt{2\pi}$ or $1/(2\pi)$ depend on conventions. Consistency between forward and inverse transforms is essential.

4. Key Takeaways & Exam Strategy Formula Card

Core Takeaway & Exam Rule

The Fourier Transform maps localized signals into frequency spectra. Tight spatial localization spreads out frequency bandwidth ($\Delta x \Delta k \geq 1/2$).

- Video reference: [Fourier Series and Transforms Part 2: Power Spectra and K-Space](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.